

ANTSIRANANA, Madagascar

WMO: 670090

Lat: 12.349S

Lon: 49.292E

Elev: 374

StdP: 14.50

Time Zone: 3.00 (E03)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	66.1	67.6	54.1	63.1	76.8	55.7	66.9	76.5	30.6	78.4	28.9	78.2	9.0	160	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	14.2	91.4	76.5	90.0	76.5	89.4	76.4	81.0	87.3	80.1	86.5	79.2	85.7	12.0	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.1	153.2	85.5	78.3	149.1	84.7	77.3	144.2	83.5	45.1	87.5	44.0	86.7	43.1	86.0	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.9	26.7	24.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	79.4	80.3	80.4	80.6	80.7	79.7	77.5	76.4	76.7	77.4	79.4	81.6	82.1	
	DBStd	2.45	1.98	1.58	1.47	1.52	1.47	1.76	1.51	1.62	1.63	1.38	1.70	1.65	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD50	10727	940	850	950	920	921	824	817	828	823	913	947	994	
	CDD65	5252	475	430	485	470	456	374	352	363	373	448	497	529	
	CDH74	43245	4111	3745	4341	4222	3681	2515	2199	2290	2523	3603	4794	5221	
	CDH80	14059	1206	1087	1398	1462	1203	722	557	593	753	1280	1857	1941	
Wind	WSAvg	12.3	6.8	5.6	6.9	9.7	14.0	14.4	15.9	18.0	18.5	15.2	13.2	8.5	
Precipitation	PrecAvg	44.7	13.0	11.9	6.7	2.4	0.6	0.8	0.8	0.8	0.4	0.5	1.9	6.5	
	PrecMax	67.4	33.5	26.5	18.7	7.1	3.5	2.9	3.0	2.1	1.3	2.2	7.6	19.4	
	PrecMin	24.6	3.3	1.4	1.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.5	
	PrecStd	13.1	7.8	5.5	3.7	1.9	0.6	0.7	0.7	0.5	0.4	0.5	1.9	5.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.6	89.8	90.9	91.5	89.6	88.2	87.5	87.7	88.2	90.1	92.8	93.0	
		MCWB	77.6	79.2	79.6	77.4	75.4	72.7	69.6	70.4	71.4	74.2	77.6	76.0	
	2%	DB	89.1	88.2	89.4	89.8	88.1	86.7	85.9	86.0	87.3	89.2	91.1	91.3	
		MCWB	78.0	78.6	78.7	77.1	74.8	71.6	69.5	70.1	71.3	73.8	75.9	76.1	
	5%	DB	87.5	87.5	88.1	88.6	87.2	85.8	84.5	84.6	86.0	87.8	89.7	89.8	
		MCWB	77.9	78.4	78.3	76.8	74.5	70.9	69.0	69.4	71.3	72.9	75.6	76.3	
	10%	DB	85.9	86.1	86.9	87.5	86.1	84.3	83.4	83.8	84.5	86.3	88.2	88.2	
		MCWB	77.6	78.3	78.0	76.4	74.1	70.2	68.8	69.1	70.7	72.1	75.0	76.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.3	81.8	81.9	80.9	79.2	75.4	74.5	74.3	76.6	77.9	81.7	81.7
			MCDB	86.7	86.9	88.1	88.6	86.5	85.8	81.8	83.6	85.0	87.3	89.0	87.5
2%		WB	79.9	80.5	80.5	79.2	77.5	73.8	72.3	72.4	74.6	76.1	79.9	79.9	
		MCDB	85.6	85.9	87.3	86.8	85.3	83.2	81.3	82.3	83.6	85.4	87.5	86.6	
5%		WB	79.0	79.5	79.3	78.3	76.2	72.7	71.1	71.3	73.2	74.9	78.4	78.8	
		MCDB	84.8	85.0	86.0	85.9	84.5	81.9	80.4	80.9	82.5	84.4	85.6	86.0	
10%		WB	78.2	78.7	78.6	77.3	75.1	71.8	70.2	70.4	72.2	73.9	77.0	77.8	
		MCDB	83.9	84.3	85.0	85.1	83.8	81.1	79.7	80.1	81.4	83.1	84.2	85.0	
Mean Daily Temperature Range	5% DB	MDBR	12.1	12.0	13.0	13.7	14.0	15.2	15.7	15.5	15.5	15.4	15.0	14.2	
		MCDBR	14.7	14.3	14.9	15.8	15.4	16.4	17.0	16.5	17.0	17.0	16.5	16.0	
		MCWBR	7.1	7.5	7.9	7.2	7.3	7.1	7.3	6.7	7.3	7.0	6.8	6.4	
	5% WB	MCDBR	13.5	13.1	13.6	14.4	14.9	15.3	15.2	15.6	15.3	15.7	15.4	14.4	
		MCWBR	7.6	7.9	8.2	7.7	8.0	7.6	7.6	7.6	8.4	7.7	8.3	7.6	
Clear-Sky Solar Irradiance	taub	0.414	0.419	0.400	0.382	0.371	0.363	0.357	0.363	0.374	0.388	0.401	0.404		
	taud	2.509	2.492	2.553	2.592	2.597	2.585	2.594	2.564	2.516	2.487	2.481	2.514		
	Ebn at Noon	295	292	291	287	280	277	282	289	295	297	297	298		
	Edh at Noon	36	37	34	31	29	29	29	31	34	36	37	36		
All-Sky Solar Radiation	RadAvg	1787	1773	1806	1798	1634	1507	1569	1752	2002	2153	2129	2000		
	RadStd	134	150	119	69	56	37	34	39	53	57	78	145		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page